

Muhammad Riyan Khan

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SUMMARY

Full stack developer specialising in Odoo ERP. I build ERP modules for HR, payroll, appraisals and finance, and the React and Flask applications that connect to them, from data model to production deployment. Outside work, I build voice and AI tools.

EXPERIENCE

Strings Technologies

Odoo Technical & Full Stack Developer

Peshawar, Pakistan

2026 – Present

Campus management system for IMSciences: Odoo 19 ERP and a self-service student portal

- Built the student portal for ~5,900 enrolled students: a React app on a Flask API gateway that reads live records from Odoo over JSON-RPC. Students check transcripts, CGPA, attendance and timetables, register for courses, and apply online for scholarships, hostel, transport and semester freeze.
- Hardened the gateway for real traffic with JWT authentication, request validation, Redis rate limiting and a caching layer that cuts repeat calls to Odoo; deployed with Gunicorn, systemd and nginx.
- Built HR and payroll for ~500 staff: promotions, transfers and disciplinary records, FBR income-tax withholding, allowances, hourly pay for visiting faculty, and payroll entries posted to the general ledger.
- Built the staff appraisal system: weighted scorecards for teaching, research and administration, approval from self-review through HOD and committee to dean, and access rules so each reviewer sees only their own department.
- Built the scholarship system and migrated 11,000+ historical awards into it from the legacy system: a ~200-field need-based application with configurable scoring, and donor-funded seats (HEC, CMEEF, FATA) that waive specific fees and settle any shortfall or surplus on the student's fee account automatically.
- Built 8 PDF and Excel reports for donor claims, fund-utilization audits and annual staff evaluations.

PROJECTS

JARVIS: voice assistant for macOS | Python, Pipecat, LLM tool calling

- A hands-free assistant that hears "Hey Jarvis", understands a request and acts on it: opening apps, controlling music and volume, setting reminders, searching the web.
- Built to run free on an 8 GB laptop: the wake word and voice run locally, and a failover layer switches between free AI providers when one hits its daily limit, so it keeps working.

Pulse: privacy-first web analytics | Next.js, Supabase, PostgreSQL

- A self-hosted alternative to Google Analytics that sites can embed with a ~1 KB script and use without a cookie consent banner.
- Counts unique visitors without cookies or stored IPs (daily-rotating salted hash); aggregation runs in PostgreSQL functions behind row-level security.

Scan2Help: emergency QR stickers | Next.js, Prisma, PostgreSQL

rayankhan.dev/projects/scan2help

- When someone has an accident, a stranger scans their sticker and gets a bilingual page with a one-tap call button to their family, with no app needed.
- Phone numbers are protected from scraping (revealed only on tap, with per-IP rate limits), and the order flow works with local payments (JazzCash/EasyPaisa).

TECHNICAL SKILLS

ERP & Backend: Odoo 19 (ORM, QWeb, access rules), Python, Flask, FastAPI, Node.js, REST, JSON-RPC

Frontend: TypeScript, JavaScript, React, Next.js, Vite, Tailwind CSS, shadcn/ui

Data: PostgreSQL (functions, row-level security), Supabase, Prisma, Redis, MongoDB

Infrastructure: Linux, nginx, systemd, Gunicorn, Vercel, Git, GitHub

AI: LLM tool calling, Pipecat, speech-to-text and text-to-speech pipelines

EDUCATION

Edwardes College

Bachelor of Science in Computer Science

Peshawar, Pakistan

2021 – 2025